AM5920 - Deep learning for Mechanics

Assignment No. 3, Due Date: 3rd May, 2026

1. Consider the following Poisson eqn

$$\frac{d^2u(x)}{dx^2} = f(x), \text{ with boundary conditions } u(x=0)=0, u(x=L)=u_L. \quad (1)$$

Consider a solution of the form

$$u(x) = \sin(2\pi x) + a\,\sin(2\pi w x) + Noise \quad (2)$$

Consider $f(x) = -4\pi^2\sin(2\pi x) - 4aw^2\pi^2\,\sin(2\pi w x)$ and two cases $a = 1/10, w(\geq 100)$ and $a = 10, w(\geq 100)$. Choose u_L as you like.

(i) Construct a physics-informed neural network using Fourier features, tanh activation function, together with the technique of Wang et al. [1] to handle gradient imbalance of the loss function for PINN, as discussed in the class, and find solution to the above PDE. Compare with exact solution to check the accuracy of the PINN solution.

(ii) Use Fourier features, tanh activation function, together with the Conditionally adaptive augmented Lagrangian method [2] for PINN to obtain results. Is it faster than than the PINN framework in question (i) to achieve similar accuracy?

(iii) Consider $L = 1, 10, 100, 1000$. Does the convergence and accuracy depend on the domain size L? If it does, what could be a simple way to make the convergence and accuracy domain size independent?

2. We are tasked to find $E(x)$ given the experimental field

$$\hat{u}(x) = \sin(2\pi x) + a_1\,\sin(20\pi x) + a_2\,\sin(200\pi x) + Noise, \quad (3)$$

for the following PDE:

$$\frac{d}{dx}\left(E(x)\frac{du(x)}{dx}\right) = 0, \text{ with boundary conditions } u(x=0)=0, u(x=L)=u_L. \quad (4)$$

Choose a value for u_L and consider three cases: $a_1 = 1/10, a_2 = 10$; $a_1 = 10, a_2 = 1/10$; $a_1 = 1, a_2 = 1$.

Choose Noise and define an appropriate loss function, which includes the mismatch between solution $u(x)$ and $\hat{u}(x)$, to infer $E(x)$.

(i) Use Fourier features, tanh activation function, together with the technique of Wang et al. [1] to handle gradient imbalance for PINN. Check convergence for mini-batch vs full-batch training.

(ii) Use Fourier features, tanh activation function, together with Augmented Lagrangian method for PINN. Check convergence for mini-batch vs full-batch training.

(iii) Can we combine Fourier features and Conditionally adaptive augmented Lagrangian method [2] to satisfy the PDE to obtain better solution? Check convergence for mini-batch vs full-batch training.

(iv) Consider $L = 1, 10, 100, 1000$. Does the convergence and accuracy depend on the domain size L? If it does, what could be a simple way to make the convergence and accuracy domain size independent?

Reference: [1] Wang, Sifan, Hanwen Wang, and Paris Perdikaris. "On the eigenvector bias of Fourier feature networks: From regression to solving multi-scale PDEs with physics-informed neural networks." Computer Methods in Applied Mechanics and Engineering 384 (2021): 113938.

[2] Hu, Qifeng, Shamsulhaq Basir, and Inanc Senocak. "Conditionally adaptive augmented lagrangian method for physics-informed learning of forward and inverse problems using artificial neural networks." Available at SSRN 5398271 (2025).